DAT321/DIT843: Quality Assurance and Testing

Questions about the exam contact:

Examiner: Francisco Gomes, francisco.gomes@cse.gu.se

Teachers: Francisco Gomes and Afonso Fontes

- Afonso will visit the exam halls around 09:30 and 10:30.
- Francisco is available for calls at the number: +46 73 856 4439
- Afonso is available for calls at the number: +46 76 564 2909

No aids are allowed for this exam.

The questions in this exam refer to the **ISO 25010:2011** that categorises four types of quality: process, internal, external and quality in use. The exam also covers the eight characteristics / attributes for product quality.

Important advice:

- Each question has points assigned shown in the square brackets. When the question is broken down into smaller sub-questions the part of the points for those specific sub-questions are also shown. Example: 1. [10 pts]. 1a. [2 pts] 1b. [8 pts].
- You must write correct, clear, readable, understandable, and unambiguous answers. If we cannot easily read what you are trying to write, then we cannot positively judge your knowledge or your learning of the topic.
- Motivate your answers (a simple statement of facts not answering the question is invalid). Whenever possible refer to modules, features and information about the software product, code or software development process related to the questions.
- Avoid circular definitions and vague / generic answers. Use unique aspects of each topic covered in your answers.
- Start each question on a new paper. Sort your answers in order (by question and sub-task) before handing them in. Write only on one side of the paper.
- Write your student code on each page and put the number of the question on every paper. The exam is anonymous, so do not leave any information that can reveal your identity.

The points and the corresponding grade are presented below (100 points in total):

- 00 – 49: U (Fail)
- 50 - 69: 3 (Pass)
- 70 - 84: 4 (Pass with credit)
- 85 - 100: 5 (Pass with distinction)

The date for the exam review session will be shared online via Canvas.

Question 1. [Total: 35 pts] You are developing an online platform called ArtSpace dedicated to the creative expressions of designers and artists worldwide (think about websites like Behance, Dribbble, DevianArt, etc.). ArtSpace offers a virtual space where creators can showcase their masterpieces such as UI designs, illustrations, paintings, photograhy, animations, etc. Artists can also use the platform to create their own designs. Below are some features and characteristics of ArtSpace:

- **F1:** Artists must create a profile to share their work. Secure login with a username and password is required to ensure account safety and personal data protection.
- **F2:** The company owning ArtSpace values quality, hence the entire infrastructure (cloud server, storage) is built such that computational resources would not be a concern for users aiming to see or display their work in the best resolution possible.
- **F3:** Artists can monetize their creations by linking a payment method to their profiles, facilitating the sale of originals, prints, or digital downloads.
- **F4:** ArtSpace allows for collaborative projects, enabling up to 20 artists to work on a single piece, sharing their progress and final works in a shared gallery space.
- **F5:** The development team values the platform's aesthetic appeal, often focusing on enhancing the user interface and experience over backend optimizations.
- **F6:** While ArtSpace is web-based for broader access, there's also a dedicated tablet app designed to cater to digital artists who prefer to create on-the-go.
- **F8:** Live streaming capabilities are integrated, allowing artists to broadcast their creative process or host live art sessions and workshops.
- **F9:** ArtSpace is designed with cross-platform compatibility in mind, ensuring artists can access the platform from any operating system, including Windows, macOS, and Linux, without any feature discrepancies.
- **F10:** ArtSpace features a dynamic recommendation engine that adapts to showcase artwork aligned with the viewer's interests and previous interactions to ensure a personalized experience.

Note: For the questions below, you are allowed to add features or decisions about the software development process (team size, technologies, etc.). However, any assumption related to your answer: (i) must be clearly state, and (ii) cannot contradict the points above.

a. [8 pts] Choose **exactly two** software product quality attributes, **and, for each,** provide (i) a brief explanation describing the quality attribute, and (ii) why they are relevant to the software product described above. When justifying your choices, you **must** refer to one or more of the points (F1 – F10) listed above.

Any two examples below should be valid options. Performance efficiency should be well justified as F2 states that there is high investment in resources, so it should not be an issue compared to other quality attributes. Arguably, the most important one will be Usability.

Usability is the degree to which the user can interact with the product. This attribute is relevant for ArtSpace because most of its features are connected to user experience elements (visualization, quality of images, resolution, etc.). F3, F4, F5, F10

Reliability is the degree to which the product performs a specific function, under specific conditions and during a period of time. This attribute is relevant for ArtSpace because the product is hosted online and used by different users simultaneously and availability of those services contributes to the professional careers of those artists. F2, F4, F8

Performance Efficiency is the degree to which the product uses computational resources. This attribute is relevant for ArtSpace because of the real-time aspect of some features where artists can share and collaborate with other artists simultaneously. F2, F4, F5, F8

Compatibility is the degree to which the product can exchange information. This attribute is relevant for ArtSpace because the product need to operate on different operating systems, or communicate with different services. F3, F8, F9

Portability is the degree to which the product can be transferred to other devices. This attribute is relevant for ArtSpace because the product needs to work on different devices such as laptops or tablets. F4, F6

Security is the degree to which the product protects information and data to control data access. This attribute is relevant for ArtSpace because the system uses authentication to access its features, and the user shares their bank information in the system. F1, F3

Maintainability is not applicable in any of the features above, hence there is not enough information to indicate its relevance. This can be acceptable, if the answer provides an assumption or context to justify it. The closest Feature that could be connected to Maintainability is F5.

b. [7 pts] For each of the quality attribute chosen above create a quality scenario for the ArtSpace system that includes: (i) source, (ii) stimulus, (iii) artefact, (iv) environment, (v) response, and (vi) response measure.

[+3 pts] per scenario, [+1 pt] for the cohesion, clarity and diversity of the scenarios.

For the quality attributes provided, here are example quality scenarios for the ArtSpace system, a platform for creative expressions akin to Behance, Dribbble, and DeviantArt:

1. Usability

- Source: End-user
- Stimulus: Click button to load portfolio of a specific artist
- Artifact: User interface in the Web browser
- Environment: Access during peak usage times.
- Response: The system presents the drawings of the artist in a grid layout.
- Response Measure: The number of columns in the grid occupies 70% of the width of the screen.

2. Portability

- Source: End-user
- Stimulus: Click button to preview one existing picture published earlier.
- Artifact: Tablet version of the Application
- Environment: The Tablet has 8,3'' inches screen. Screen is in portrait mode. Image is a horizontal painting.
- Response: The screen shows the image without any rotation.
- Response Measure: Image shown fits 100% of the device's screen width in the portrait mode.

3. Security

- Source: End-user
- Stimulus: Logins into the platform
- Artifact: Login screen and authentication component.
- Environment: User is registered and password is incorrect. There were 5 attempts with incorrect login.
- Response: The system reloads the screen and shows a message "Too many unsuccessful login attempts. To recover your account, access the link sent to your e-mail.".
- Response Measure: Temporarily locks the account for 5 minutes to prevent brute force attacks.

c. [12 pts] You are asked to analyse external quality and quality in use of ArtSpace. (i) Explain the difference between those two types, (ii) describe one measure / metric for each type, and (iii) illustrate how that measure can be used to assess the quality of one or more of the features described above.

[4 pts] Describe the difference: External quality focuses on the execution of the system (i.e., its behaviour) and is part of the product quality, whereas quality in use focuses on the end-user's perspective where the product is evaluated under the view of the end-user and in a specific context of usage.

[4 pts] Any measure for reliability, latency, usability, or security is acceptable for External quality, such as Availability (F4, F5, F8), latency (F2, F8), jittering, number of hacking attempts (F1, F3), number of installations on MacOS (F9), or tablet (F6), etc.

[4pts] For Quality in Use, measures that convey the end-user's engagement and positive/negative response to the product are applicable, such as user satisfaction (e.g., measured by Likert scale -), or increase / decrease in number of subscribers (F1, F3), or financial transactions to buy different artists' work (F3), users working simultaneously (F4), users watching streams (or liking the streams) (F8).

d. [8 pts] ArtSpace wants to promote sustainability in the product that we are creating. Using the Sustainability Awareness Framework, you must (i) choose or suggest one functionality for the product and explain the (ii) immediate, (iii) enabling and (iv) structural effects of those features. You must also indicate the corresponding dimension of sustainability in which those effects.

[+2 pts] for correct immediate effect description.
[+2 pts] for correct enabling effect description.
[+2 pts] for correct structural effect decription.
[+2pts] for correct dimensions in all effects.

A few examples are shown below.

Example 1:
ArtSpace will promote the use of sustainable materials by artists and designers, that can tag that they used sustainable products in the creation of their shared art. This is done via a "tag" that can be added to the uploaded material.

Immediate effect: This increases awareness and informs viewers about sustainable practices in art creation. (Environmental dimension)

Enabling effect: This feature enables better searchability for environmentally-friendly art, leading to higher visibility and potentially more sales for artworks that use sustainable practices. (Economic, or Environmental)

Structural effect: Over time, as the feature becomes widely used, it could lead to a structural shift in the art industry, with suppliers producing more sustainable materials to meet demand. (Economic, or Environmental)

Example 2:
ArtSpace will create a specific area within the platform dedicated to showcasing and fostering collaboration among artists from minority groups.

Immediate effect: Artists from minority groups would have a platform to express their cultural heritage and personal experiences, which can raise awareness and appreciation for diverse perspectives in art. (Individual or Social dimension)

Enabling effect: This space could enable minority artists to gain greater visibility and opportunities, potentially leading to more collaboration and cross-cultural exchanges. (Social dimension)

Structural effect: In the long term, the existence of such a space could influence the broader art world to be more inclusive and representative of minority groups. This could also include more funding opportunities for minority artists, and a re-evaluation of the criteria for success and recognition in the art industry. (Social or Economic dimensions)

Question 2. [Total: 35 pts] You are asked to test the program specified below. You are also given the list of functions below to assist your process of writing test cases, or properties for your software. Based on your knowledge about Software Testing Techniques, answer the questions below.

Specification: The function `mergeInMiddle(listA,listB)` combines the elements of two lists in a single list. The method receives as input two lists of integers: listA, listB. The output is a single list where all elements of listB are inserted in the middle index of listA. If any of the lists is null, the method will throw an Exception. See the test cases below:

Note: Since the implementation is done in Java, you can assume that: (i) there cannot be null elements inside the list of integers, (ii) the first index of the list is 0, and (iii) the middle index is: `floor(listA.length/2)`

T1	`mergeInMiddle([1,0,1],[10,10,5]) → [1, 10, 10, 5, 0, 1]`
T2	`mergeInMiddle([1,2,3,4],[2,1]) → [1, 2, 2, 1, 3, 4]`
T3	`mergeInMiddle([],[2,1]) → [2,1]`
T4	`mergeInMiddle([-5, -10],[]) → [-5, -10]`
T5	`mergeInMiddle([],[]) → []`
T6	`mergeInMiddle([92, 42], null) → Exception`
T7	`mergeInMiddle([100], [50, 60, 10]) → [50, 60, 10, 100]`

List of methods that can be used to write your properties:

`sort(int[]): int[]`	Returns the array with its values sorted in an ascending order.
`reverse(int[]):int[]`	Returns the array where its values are in reversed positions.
`length(int[]): int`	Returns the number of elements in the array.
`sum(int[]): int`	Returns the sum of elements in the array.

`max(int[]):int`	Returns the highest value inside the array
`min(int[]):int`	Returns the lowest value inside the array

a. [15 pts] Create exactly five properties to test the `mergeInMiddle` function. You can write the properties using natural language, mathematical notation, or pseudocode. Feel free to use the support functions listed above. Your properties will be evaluated based on the diversity of their input and output.

[3 pts] per property. The properties should be diverse, i.e., some could use the support functions listed above, and some could not. Some examples below:

```
Using the SUT and properties of the arrays:
1. For any, a, b = []: middle(a, b) == middle(b, a) == a
2. For any a, middle(null, a) == middle(a, null) == Exception
3. For any a,b, b != []; and mIndex = (length(a) + length(b)) / 2
      a. b[0] == middle(a,b)[mIndex]
4. For any a,b a != []: a[0] == middle(a,b)[0]

Using the support functions:
5. For any a, b: max(max(a), max(b)) == max(middle(a,b)
6. For any a, b: min(min(a), min(b)) == min(middle(a,b)
7. For any a,b: length(a) + length(b) == length(middle(a,b))
8. For any a,b: sum(a) + sum(b) == sum(middle(a,b))
```

b. [7 pts] Using your knowledge of test oracles, (i) explain two different types of oracles that can be used to generate test cases. For each property that you suggested, (ii) state and justify which type of oracle helped you create it.

[4 pts] for the explanation of the oracles, and [3 pts] for the correct classification.

Specified oracles are those where the tester rely on the system's requirements, documents or other specification artefacts.

Human oracles are those where the tester rely on their expertise, experience, or domain knowledge to derive tests.

Implicit oracles are those where the tester uses general expectations or implicit knowledge about the system's behaviour, such as overflow errors, divisions by zero, etc.

Derived oracles are those where the tester uses information created from different software artefacts that are not its specification (e.g., build logs, past execution traces, etc.)

The SUT described above is not domain specific (e.g., it is not related to an application such as Spotify, Twitter, Text Editor, etc.), so it is difficult to have a human oracle. Most oracles will be either implicit or specified. For the examples above: Properties 1—4 are all specified oracles, whereas properties 5—8 are implicit oracles (derived oracles could also be acceptable).

c. [8 pts] In the example of test cases above, your colleague Charlie used equivalence class partitioning and argues that T3, T4, and T5 belong to the same partition. Do you agree or disagree with Charlie? (i) Explain what equivalent class partitioning is, and (ii) justify why you agree/disagree with Charlie.

Agree with Charlie.

[3 pts] Equivalence partitioning is a black-box (or specification-based) technique where we divide the input space into sets of input that trigger similar behaviour in the software. These sets of input cannot have overlapping elements.

[5 pts] I agree with Chalie. The partition class here are those sets of inputs (pairs of arrays) where one of them is empty. Note that in T3, T4 and T5 at least one of the arrays is empty. The SUT behaviour is that the output will be equal to the "other" array that (typically non-empty but also applies to two empty arrays). All three test cases cover that set of inputs, hence they belong to the same partition.

d. [5 pts] There are numerous criteria used to check the quality of a test suite. Choose one test adequacy criteria, explain it, and mention one advantage and one disadvantage of that criteria.

Here multiple options apply: e.g., mutation coverage, statement coverage, MCDC coverage, test diversity, etc. The typical answer would include either mutation coverage or one of the code coverage measures. What is not acceptable is to simply state "code coverage" without specifying the specific code coverage used. Whichever measure chosen, the answer should include:

[+1 pts] explanation, [+2 pts] advantage, [+2 pts] disadvantage.

The answers should focus on the unique aspect of the criteria. For instance, branch, statement or any of the code coverage, do not care about the specific input value, such that some boundary values could be missed. In mutation, the main disadvantage is inefficiency of the mutation tools, or the risks of having equivalent mutants reducing the mutation scores.

Question 3. [Total: 30 pts] You are responsible for testing a software component developed by a third-party. Since the component is confidential, you do not have access to its code or specification. However, your collaborator has agreed to share with you the diversity of five test cases (TC) created by their own engineers. Based on your knowledge of diversity-based testing and regression testing, answer:

Input Distance

	TC1	TC2	TC3	TC4	TC5
TC1	-	0.15	0.15	0.45	0.18
TC2		-	0.05	0.55	0.33
TC3			-	0.50	0.14
TC4				-	0.80
TC5					-

Output Distance

	TC1	TC2	TC3	TC4	TC5
TC1	-	0.80	0.15	0.67	0.40
TC2		-	0.18	0.65	0.33
TC3			-	0.43	0.25
TC4				-	0.90
TC5					-

The two matrices above show the information about the diversity between pairs of tests in a test suite. The matrix to the left was obtained by using distance functions in the input value of the test case, whereas the matrix to the right is obtained from applying the same distance function but in the expected output of the SUT.

a. [8 pts] Using the information provided, identify one boundary in the System Under Test. Justify your answer.

[5 pts] We can use the intuition of the program derivative which identifyies boundaries by finding pairs of tests that have a very similar input (low input distance) and produce very different outputs (high input distance).

[3 pts] The boundary is between T1 and T2.

b. [7 pts] List one advantage and one disadvantage of using diversity-based testing for regression testing.

[+3 pts] for each (adv. and disad.) and [+1 pt] for clarity and cohesion of the written answer.

Some examples of advantages (pick one):

- Diversity can be calculated on any type of data (strings, numbers, images, bytes, etc.)
- Diversity is useful when there is no prior knowledge about the domain or the SUT itself, because the diversity information is embedded in the artefacts.
- By covering diverse tests, we increase the coverage of the input space, hence increasing our chances of finding bugs "hidden" within the input space.

Some examples of disadvantages (pick one):

- Diversity can be very slow due to the pairwise comparison of test, and calculation of distance functions.
- Diversity-based regression testing, like any other type of cost-effective regression testing, can skip faults when executing fewer tests.

c. [10 pts] Your colleagues Charlie and Ellis have used the output diversity to prioritise two different test suites to be sent to your collaborator. The tests in the first indices (left) are executed first.

- Charlie: [TC4, TC5, TC2, TC1, TC3]
- Ellis: [TC3, TC1, TC5, TC4, TC2]

Which test suite above is better prioritized for output diversity? Justify your answer.

[2pts] State which one is best: The best test suite for output diversity is Charlie's.

[4 pts] Describe the reasoning behind it: We should choose pairs of tests that have higher output distance values in the matrix.

[4 pts] Describe the step-by-step: Looking at the matrix to the right, T4 and T5 have the highest distance value, hence should be executed first. The next step is choosing the highest pair that is different than the remaining tests, but also, different to the already chosen ones. T2 and T1 are different from each other (second highest value in the matrix), but also different from T4 and T5.

Some variations of the explanations above could include the algorithms describes in lectures, or a process of exclusion by identifying that Ellis' test suite starts with the most similar tests (lowest value from the matrices).

d. [5 pts] Your colleague Ellis suggested using mutation testing to compare if the prioritised test suite is better than the original test suite. Do you agree with Ellis's suggestion? Justify your answer.

Disagree with Ellis. Note that prioritisation alone does not reduce the set of tests, it simply assigns an order of the tests. Therefore, the prioritised test suite per se includes the same tests as the original test suite, therefore, whatever mutation score is obtained in the original suite, will also be obtained by the prioritised test suite.